

Nitrogen-Doped TiO₂ Nanotubes-Catalyzed Synthesis of Small D- π -A-Type Knoevenagel Adducts and β -Enaminones

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For the first time, interstitial nitrogen-doped TiO₂ nanotubes (INDTNTs) have been utilized as heterogeneous green catalysts for the synthesis of D- π -A-type Knoevenagel adducts (**3**) and β -enaminones (**6**). Notable advantages of the present method include its wide substrate scope, low catalyst loading, avoiding the use of hazardous bases and solvents, products do not require chromatographic purification, high yields, catalyst reusability, etc. The optimized catalyst, INDTNTs-4 has been characterized by XRD, HRTEM, and XPS techniques. The photo-

physical and electrochemical properties of D- π -A-type **3** and **6** have been evaluated. The study revealed that the colour tunability (cold white light to orange light) and the HOMO/LUMO levels depend mainly on the presence of various electron donating and withdrawing groups on the aromatic/heteroaromatic rings of **3** and **6**. The results indicate that the D- π -A-type **3** and **6** could be promising materials for the fabrication of optoelectronic devices.

Introduction

In recent years, π -conjugated small organic semiconductor materials (OSCMs) have received substantial interest because of their promising applications in a variety of research fields associated with light like field effect transistors, memory devices, sensors, solid-state lasers, solar cells, photovoltaics, light emitting diodes, etc.^[1–9] OSCMs have several unique features as compared with inorganic semiconductor materials: ease of preparation, low temperature processing, high luminescence efficiency, mechanical flexibility, variable optical band gaps, low cost, mercury-free, light weight, easy to tune and control of colours, etc.^[10–17] The renowned electronics companies affording OLED based displays in their new products such as mobile phones, UHDTVs, etc.^[18,19] An OLED device is constructed by using a hole-transport layer (HTL), organic emissive layer (OEL) and electron-transport layer (ETL) which are inserted between cathode and anode.^[20–22] The structure, stability, charge mobility and molecular orbital (highest occupied molecular orbital (HOMO) and lowest unoccupied molec-

ular orbital (LUMO)) levels of each layer have played a crucial role in design and development of highly efficient OLEDs. Hence, the search for identifying the small π -conjugated OSCMs, which are able to emit visible light efficiently in the solid state is the primary objective in the modern OLEDs research. For example, Knoevenagel adducts and β -enaminones are considered as small D- π -A-type molecular architectures.

Carbon-carbon double bond forming reactions like Knoevenagel condensation and β -enaminones formation are important transformations in synthetic organic chemistry.^[23] The products obtained in these conversions are widely used in the various fields of chemistry. Knoevenagel adducts are utilized as starting materials in the synthesis of fine chemicals, pharmaceutical drugs, natural products, perfumes, agrochemicals, cosmetics, herbicides, insecticides, functional polymers, etc.^[24–29] Due to their potential utility, various homogeneous and heterogeneous catalysts like amines,^[30] alkali salts,^[31] ionic liquids,^[32] self-assembled solids,^[33] polyoxometalates,^[34] metal oxides,^[35] immobilized mesoporous silica,^[36] Al₂O₃,^[37] polyacrylamide,^[38] polyacrylonitrile fiber,^[39] mesoporous C₃N₄^[40] and zeolites^[41,42] have been utilized for Knoevenagel condensation. Despite the efficiency of above-mentioned catalysts, still there is a scope for the development of green and sustainable heterogeneous catalytic systems for Knoevenagel and other related condensation reactions.

β -Enaminones are versatile synthons in organic synthesis because they have both nucleophilic 'enamine' and electrophilic 'enone' functions. Due to this amphoteric nature, nucleophilic 'enamine' character of β -enaminones is used as a valuable synthon in the preparation of different heterocyclic scaffolds. 'Enone' character of these compounds can act as electrophilic center for both 1,2- and 1,4-additions.^[23d] Hence, the β -enaminones are employed as key starting materials for the synthesis of various heterocycles and biologically active compounds.^[43–45] Inspired by the importance of this valuable scaffold, various synthetic routes have been developed for the

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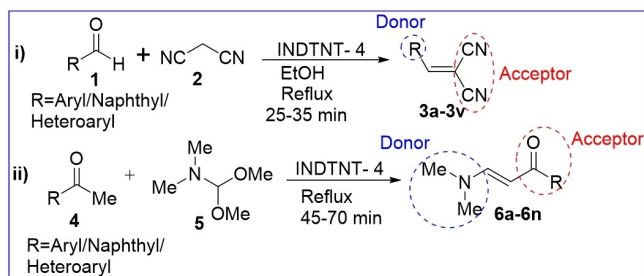
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synthesis of β -enaminones.^[46–52] Some of the reported methods suffer from disadvantages, such as the use of an excess amount of DMF-DMA, employs expensive and hazardous catalysts and solvents, lengthy reaction times, high temperatures, etc. Hence, there is an urgent need to develop a green and sustainable synthetic method which addresses the above problems in the synthesis of β -enaminones.

Over the past few decades, the impact of nanocatalysis is augmented not only in organic chemical industry but also in academia because this catalysis meets most of the green chemistry principles. Nanocatalysts are employed in a wide range of organic transformations, for example carbon-carbon or carbon-heteroatom bond forming reactions and these catalysts outwardly operate under heterogeneous conditions. Transition metal oxides, particularly, TiO_2 nano materials occupied an important place as reusable heterogeneous catalyst in various organic transformations like Friedel–Crafts alkylation, Beckmann rearrangement, etc.^[53] This is due to their distinctive properties which include its morphology, high specific surface area, mixed-phase composition, good thermal and mechanical stability, etc. Further, nitrogen doping TiO_2 nano materials received substantial attention because of their promising photo catalytic activities.^[54] But not much is explored in organic transformations as catalysts. Extensive literature study reveals that the N-doped TiO_2 nanorods are used as heterogeneous catalyst for Kabachnik–Fields reaction under microwave irradiation.^[55] No research activity has been reported on interstitial N-doped TiO_2 nanotubes (INDTNTs) catalyzed carbon–carbon double bond forming reactions, Knoevenagel condensation and β -enaminones formation.

Herein, the use of INDTNTs as heterogeneous catalysts is described for the following carbon–carbon double bond forming reactions: i) Knoevenagel adducts (**3**) from various aromatic/heteroaromatic aldehydes (**1**) and malononitrile (**2**) and ii) β -enaminones (**6**) from aromatic/heteroaromatic methyl ketones (**4**) and N,N-Dimethylformamide dimethyl acetal (DMF-DMA) (**5**) (Scheme 1). The photophysical and electrochemical properties of the prepared small D- π -A-type **3** and **6** have also been investigated.



Scheme 1. INDTNTs-4 catalyzed Knoevenagel condensation and β -enaminones formation.

Results and Discussion

Initially, TiO_2 nanotubes (TNTs) and various INDTNTs have been synthesized by adopting previously reported methods.^[48,50,51] Firstly, TNTs were fabricated by hydrothermal method from anatase- TiO_2 . A series of INDTNTs (INDTNTs-1 to INDTNTs-4) were produced from TNTs and β -alanine (as mentioned in Table 1) through a solid-state dispersion method.

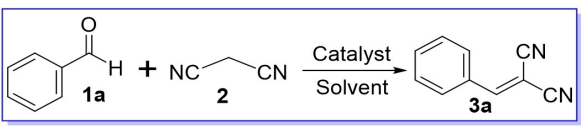
The prepared INDTNT catalysts were employed for the construction of carbon–carbon double bond forming condensation reactions like Knoevenagel condensation and β -enaminone formation. Firstly, the Knoevenagel reaction conditions were optimized using benzaldehyde (**1a**) and malononitrile (**2**) as benchmark substrates. The control reaction was performed in water at room temperature for 3 h. But the formation of **3a** was not noticed. The same reaction was also performed in various solvents like acetone, isopropanol, EtOH and acetonitrile; the yields of **3a** were 0, 5, 20 and 0%, respectively (entry 1, Table 2). Commercially available TiO_2 and the prepared catalysts, TNTs, INDTNTs-1, INDTNTs-2, INDTNTs-3 and INDTNTs-4 in various solvents were examined for the improvement of the yield of **3a** (entries 2–7, Table 2). The study revealed that 10 mg of INDTNTs-4 in EtOH at reflux temperature afforded good yield (85%) of product (**3a**) (entry 7/EtOH, Table 2). INDTNTs-1, INDTNTs-2 and INDTNTs-3 in ethanol produced moderate to good yields of **3a** (entries 4–6/EtOH, Table 2). TiO_2 and TNT gave very low yields of **3a** (entries 2 and 3/EtOH, Table 2). Further, the amount of INDTNTs-4 was increased (15, 20, 25, 30 and 35 mg) for the better yields of **3a** (Table 3). The study revealed that the INDTNTs-4 (20 mg) afforded 98% of **3a** in 30 min at reflux temperature in EtOH (entry 3, Table 3). The yields of **3a** did not increase even when the amount of INDTNTs-4 was increased (entries 4–6, Table 3).

Under best-yielding reaction conditions for the Knoevenagel reaction, the scope of various functionalized aldehydes and malononitrile was investigated. As can be seen from Table 4, benzaldehyde containing electron donating groups (EDGs), 4- OCH_3 (**1b**), 3-hydroxy (**1c**), 4-hydroxy (**1d**), 4- $\text{N}(\text{CH}_3)_2$ (**1e**), 4- OPh (**1f**), 3,4-dihydroxy (**1g**), 2,4-dimethoxy (**1h**), 3-ethoxy-4-hydroxy (**1i**) and 3,4,5-trimethoxy (**1j**) groups exhibited good reactivity with malononitrile (**2**) in the presence of INDTNT-4 to yield the respective adducts **3b–3j** in excellent yields ranging from 94 to 98%. The absolute structure of compound **3b** has been confirmed by single crystal X-ray diffraction technique (Figure S31 (see Supporting Information)). Benzaldehyde having electron withdrawing groups (EWGs), 2- NO_2 (**1k**), 4-F (**1l**) and 4-Cl (**1m**) groups reacted well with malononitrile (**2**) in the presence of INDTNTs-4 to provide the corresponding products

Table 1. Chemical composition for the preparation of various INDTNT catalysts.

S.No.	Catalyst code	Wt. of TNT [g]	Wt. of β -Alanine [g]	Ratio
1	INDTNTs-1	0.5	0.25	1:0.5
2	INDTNTs-2	0.5	0.5	1:1
3	INDTNTs-3	0.5	0.75	1:1.5
4	INDTNTs-4	0.5	1.0	1:2

Table 2. Screening of reaction conditions.^[a]

					
Entry	Catalyst	Solvent (5 mL)	t [min]	Product	Yield ^[b] [%]
1 ^[c]	–	H ₂ O	180	3a	–
		Acetone	180		–
		Isopropanol	180		05
		EtOH	180		20
2	TiO ₂	Acetonitrile	180	3a	–
		H ₂ O	60		10
		Acetone	60		–
		Isopropanol	60		15
		EtOH	60		20
3	TNT	Acetonitrile	60	3a	–
		H ₂ O	60		10
		Acetone	60		–
		Isopropanol	60		15
		EtOH	60		25
4	INDNTs-1	Acetonitrile	60	3a	–
		H ₂ O	60		25
		Acetone	60		–
		Isopropanol	60		35
		EtOH	60		50
5	INDTNTs-2	Acetonitrile	60	3a	20
		H ₂ O	60		25
		Acetone	60		–
		Isopropanol	60		40
		EtOH	60		65
6	INDTNTs-3	Acetonitrile	60	3a	25
		H ₂ O	60		30
		Acetone	60		–
		Isopropanol	60		50
		EtOH	60		76
7	INDTNTs-4	Acetonitrile	60	3a	30
		H ₂ O	60		40
		Acetone	60		–
		Isopropanol	60		65
		EtOH	60		85
		Acetonitrile	60		35

[a] Conditions: Benzaldehyde (**1a**) (5 mmol) and malononitrile (**2**) (5 mmol), catalyst (10 mg), solvent (5 mL), reflux temperature. [b] Yield. [c] Reaction in the absence of catalyst.

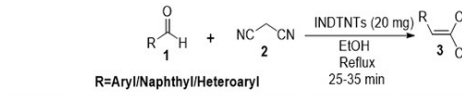
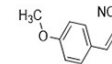
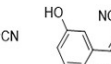
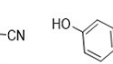
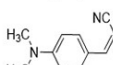
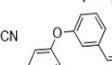
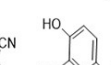
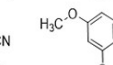
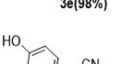
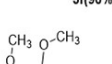
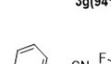
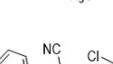
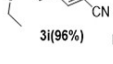
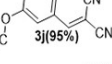
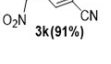
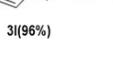
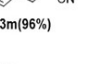
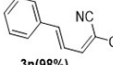
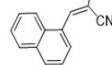
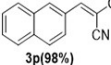
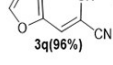
Table 3. Loading of INDNTs-4 catalyst.^[a]

Entry	INDTNTs-4 [mg]	Product	t [min]	Yield ^[b] [%]
1	10	3a	60	85
2	15	3a	30	91
3	20	3a	30	98
4	25	3a	30	98
5	30	3a	30	98
6	35	3a	30	98

[a] Conditions: Benzaldehyde (**1a**) (5 mmol) and malononitrile (**2**) (5 mmol), INDNTNTs-4 catalyst, EtOH (5 mL), reflux temperature. [b] Yield.

3k–3m in good isolated yields (91 to 96%). Cinnamaldehyde (**1n**), 1-naphthaldehyde (**1o**) and 2-naphthaldehyde (**1p**) also afforded **3n**, **3o** and **3p** in excellent isolated yields. Next, a variety of heteroaromatic aldehydes, furan-2-aldehyde (**1q**), thiophene-2-aldehyde (**1r**), 4-methylthiazole-5-carbaldehyde (**1s**), 2-aminonicotinaldehyde (**1t**), 1H-indole-3-carbaldehyde

Table 4. Substrate scope of aromatic/heteroaromatic aldehydes.^[a]

	
R = Aryl/Naphthyl/Heteroaryl	
	
	
	
	
	
	
	
	
	
	
	
	

[a] Conditions: Aromatic/heteroaromatic aldehydes (**1**) (5.0 mmol) and malononitrile (**2**) (5.0 mmol), INDNTNTs-4 (20 mg), EtOH (5.0 mL), reflux temperature.

(**1u**) and 2-chloro-8-methylquinoline-3-carboxaldehyde (**1v**) showed good reactivity with malononitrile (**2**) in the presence of INDNTNTs-4 to give the corresponding Knoevenagel adducts, **3q**, **3r**, **3s**, **3t**, **3u** and **3v** in good isolated yields.

Encouraged by the above results, the catalytic efficiency of INDNTNTs-4 was also studied in the synthesis of β -enaminones (**6**) which were used as versatile synthons for the synthesis of medicinally privileged heterocyclic scaffolds. For this purpose, firstly, a control experiment was conducted by employing the model substrates, acetophenone (**4a**) and DMF-DMA (**5**) for the preparation of **6a** under solvent and catalyst free conditions at reflux temperature for 120 min. It was noticed that the **6a** was not formed. Next the same reaction was performed in the presence of INDNTNT-4 catalyst (20 mg) at reflux temperature for 45 min under solvent free conditions, which resulted in 98 % yield of **6a**.

With the best green reaction conditions in hand, we explored the substrate scope of this method using a variety of aromatic/heteroaromatic methyl ketones. As can be seen from Table 5, acetophenone bearing EDGs, 4-CH₃ (**4b**), 4-OCH₃ (**4c**), 4-OH (**4d**) and 4-NH₂ (**4e**) groups showed excellent reactivity with DMF-DMA (**5**) in the presence of INDNTNTs-4 to give the corresponding β -enaminones (**6b–6e**) in good yields ranging from 96 to 98 %. Acetophenone containing EWGs, 4-NO₂ (**4f**), 4-CN (**4g**), 4-Cl (**4h**), 4-Br (**4i**) and 2,4-dichloro (**4j**) groups, reacted well with DMF-DMA (**5**) in the presence of INDNTNTs-4 to afford the products **6f–6j** in excellent isolated yields (93 to 97 %). 2-Acetylnaphthalene (**4k**) also provided **6k** in good isolated yield. Further, heteroaromatic methyl ketones, 2-acetylpyridine (**4l**), 2-

Table 5. Substrate scope of aromatic/heteroaromatic methyl ketones.^[a]

$\text{R} \begin{array}{c} \text{O} \\ \parallel \\ \text{C} \\ \\ \text{Me} \end{array} + \text{Me} \begin{array}{c} \text{Me} \\ \\ \text{N} \begin{array}{c} \text{OMe} \\ \\ \text{OMe} \end{array} \end{array} \xrightarrow[\text{45-70 min}]{\text{INDTNTs-4 (20 mg), Reflux}} \text{Me} \begin{array}{c} \text{O} \\ \parallel \\ \text{C} \\ \\ \text{N} \begin{array}{c} \text{OMe} \\ \\ \text{OMe} \end{array} \end{array} \text{R}$ $\text{R} = \text{Aryl/Naphthyl/Heteroaryl}$	
	6a (98%)
	6b (97%)
	6c (98%)
	6d (96%)
	6e (98%)
	6f (96%)
	6g (97%)
	6h (97%)
	6i (97%)
	6j (93%)
	6k (97%)
	6l (92%)
	6m (95%)
	6n (90%)

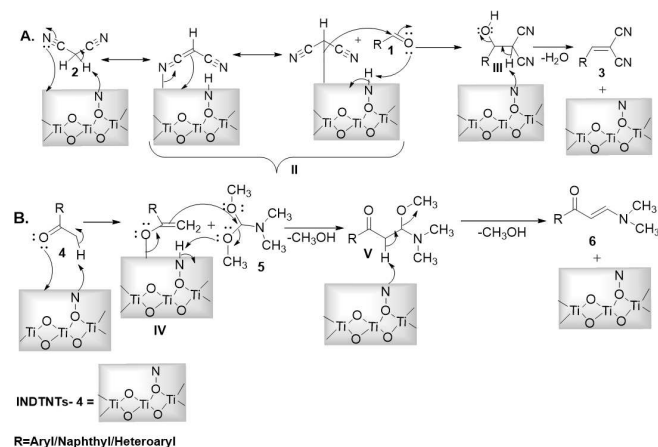
[a] Conditions: Aromatic/heteroaromatic methyl ketones (4) (5.0 mmol) and DMF-DMA (5) (5.0 mmol), INDTNTs-4 (20 mg), reflux temperature.

acetylthiophene (**4m**) and 3-acetylcoumarin (**4n**) produced the corresponding β -enaminones, **6l**, **6m** and **6n** in 92%, 95% and 90% yields, respectively.

Mechanism

The mechanism for the formation of Knoevenagel adducts (**3**) and β -enaminones (**6**) catalyzed by INDTNTs-4 has been described in Scheme 2.

A. Knoevenagel condensation starts with abstraction of proton from the malononitrile (**2**) by INDTNTs-4 to obtain a resonance stabilized titanium complex of malononitrile (**II**). Next, the nucleophilic addition reaction between intermediate (**II**) and the aromatic aldehydes (**1**) to form 2-(1-hydroxyaryl/naphthyl/heteroaryl) malononitrile (**III**). Subsequently, intermediate (**III**) undergo dehydration in the presence of INDTNTs-4 yields the Knoevenagel adducts (**3**) (Scheme 2A).



Scheme 2. Plausible mechanism for the formation of A) Knoevenagel adducts (**3**) and B) β -enaminones (**6**) catalyzed by INDTNTs-4.

B. At first, INDTNTs-4 removes a proton from the methyl ketone (**4**) to form a titanium complex of enolate ion (**IV**). Nucleophilic addition of enolate ion (**IV**) to an acetal carbon of DMF-DMA (**5**) to give an intermediate, 3-(dimethylamino)-3-methoxy-1-aryl/naphthyl/heteroarylpropan-1-one (**V**). Subsequently, intermediate (**V**) undergoes elimination of methanol in the presence of INDTNTs-4 to obtain the desired β -enaminones (**6**) (Scheme 2B).

Reusability

The reusability of INDTNTs-4 was then studied in the synthesis of **3b** using **1b** and **2** under the established reaction conditions. After the first cycle, the catalyst was separated by filtration and dried. To know the catalytic efficiency, the recovered catalyst was reused in the same model reaction for 07 times and the yields obtained of **3b** were 98, 98, 98, 97, 97, 97 and 97% for 1st, 2nd, 3rd, 4th, 5th, 6th and 7th cycles, respectively. The study revealed that the catalytic efficiency of the recovered INDTNTs-4 is similar to the fresh catalyst.

Characterization of INDTNTs-4 catalyst:

The optimized catalyst (INDTNTs-4) has been characterized for its structure, morphology and elemental analysis through XRD, HRTEM with energy dispersive spectroscopy (EDS) and XPS techniques.

X-ray diffraction

Figure 1 explains the powder XRD pattern of INDTNTs-4 catalyst. The diffraction peaks at $2\theta = 22.80, 25.32, 37.12, 37.92, 38.69, 43.27, 48.02, 53.94, 55.20, 62.77, 68.87, 70.23, 75.08$ and 82.86 were attributed to the (101), (004), (200), (105), (211), (204), (116), (120) and (215) planes of anatase TiO_2 (JCPDS 01-084-1286). Further, it was observed that the broad nature of the XRD of INDTNTs-4 catalyst indicated that the ceramic characteristics of the catalyst.

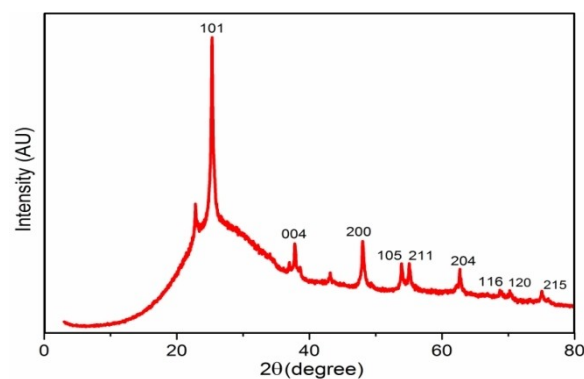


Figure 1. XRD pattern of INDTNTs-4 catalyst.

High-resolution transmission electron microscopy

HRTEM images of INDNTs-4 are shown in Figure 2a–b. As can be seen from Figure 2a–b, it was observed that the nitrogen was deposited on the surface of TiO₂ nanotubes. The lighter appearance of these nanotubes was owing to the lesser thickness of INDNTs-4. Nevertheless, the overall morphology of INDNTs-4 exhibits nanotubes in nature with different sizes and orientations. The elemental composition of INDNTs-4 was examined by using EDS technique. From the spectrum, it was noticed that the presence of N, O and Ti elements in INDNTs-4 (Figure 2c). The appearance of Cu peak is originated from the grid used for recording EDS spectrum.

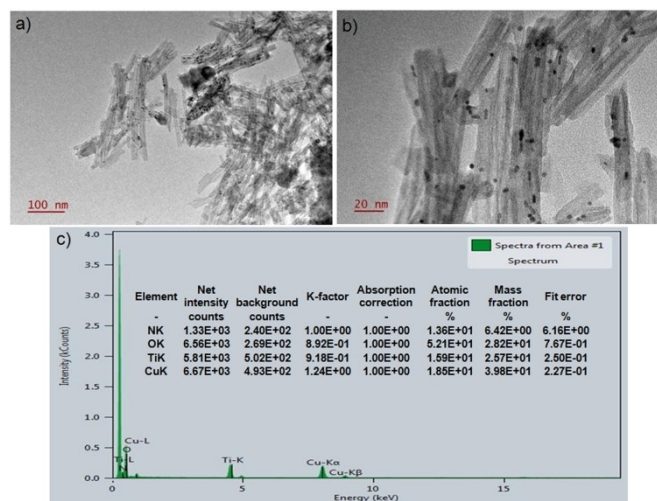


Figure 2. (a)–(b) HRTEM and (c) EDS spectrum of INDNTs-4.

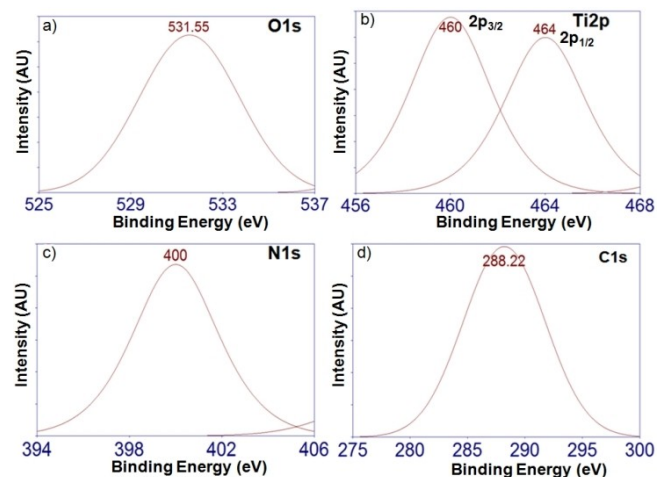


Figure 3. XPS spectra of INDNTs-4 catalyst (a) O1s; (b) Ti2p; (c) N1s; (d) C1s.

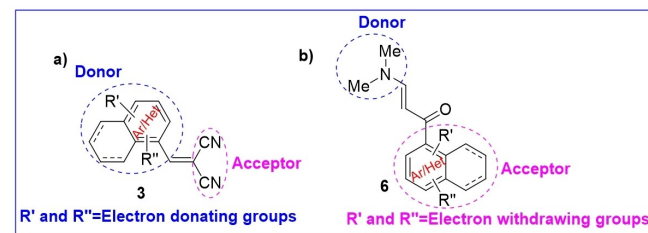
X-ray photoelectron spectroscopy

The electronic states of elements in INDNTs-4 were measured by XPS spectra (Figure 3). A peak centered at 531.55 eV in the O1s spectrum indicates the presence of O1s in the INDNTs-4 catalyst (Figure 3a). The Ti 2p spectrum has (Figure 3b) two prominent bands centered at 460 and 464 eV corresponding to Ti 2p_{3/2} and Ti 2p_{1/2} of TiO₂. N1s spectrum displayed a peak centered at 400 eV which was ascribed to interstitial N in the INDNTs-4 (Figure 3c).^[56a–c] Further, a binding energy shift to higher values compared to the usual appearance of N1s (~397.5 eV) indicate the presence of oxidized nitrogen owing to the formation of Ti–O–N linkages.^[56d] A peak centered at 288.22 eV in the C1s spectrum confirmed the presence of carbon in the INDNTs-4 catalyst (Figure 3d).

Photophysical properties of Knoevenagel adducts (3) and β -enaminones (6)

Photophysical properties of a library of synthesized D- π -A-type simple Knoevenagel adducts (3) and β -enaminones (6) (Scheme 3) were evaluated from their absorption spectra, luminescence spectra and decay curves. The obtained results are discussed hereunder.

From the optical absorption and emission spectra of Knoevenagel adducts (3) (Figure S32 (see Supporting Information) and Figure 4), it is found that the absorption and emission of 3 are influenced by the substituents on aromatic/hetero-aromatic ring. For instance, the electron donating -N(CH₃)₂ group at 4th position of the phenyl ring of 3e (4-N(CH₃)₂, λ_{em} = 630 nm) was bathochromically shifted with a small Stokes shift (7437 cm⁻¹) as compared to the other electron donating substituents at different positions of the phenyl ring, i.e. 3b (4-OMe, λ_{em} = 503 nm, Stokes shift = 9274 cm⁻¹), 3c (3-OH, λ_{em} = 507 nm, Stokes shift = 13279 cm⁻¹), 3d (4-OH, λ_{em} = 522 nm, Stokes shift = 11054 cm⁻¹), 3h (2,4-diOMe, λ_{em} = 553 nm, Stokes shift = 8513 cm⁻¹), 3i (3-ethoxy-4-hydroxy, λ_{em} = 529 nm, Stokes shift = 8344 cm⁻¹) and 3j (3,4,5-triOMe, λ_{em} = 561 nm, Stokes shift = 10344 cm⁻¹). It was also noted that the compound, 3e (4-N(CH₃)₂, λ_{em} = 630 nm; Stokes shift = 7437 cm⁻¹) was red shifted with a large Stokes shift when compared to 3g (3,4-dihydroxy, λ_{em} = 525 nm, Stokes shift = 2979 cm⁻¹). The compound, 3p (β -naphthyl group, λ_{em} = 585 nm, Stokes shift = 10303 cm⁻¹) was slightly bathochromically shifted with a small



Scheme 3. Schematic representation of D- π -A-type a) Knoevenagel adducts (3) and b) β -enaminones (6).

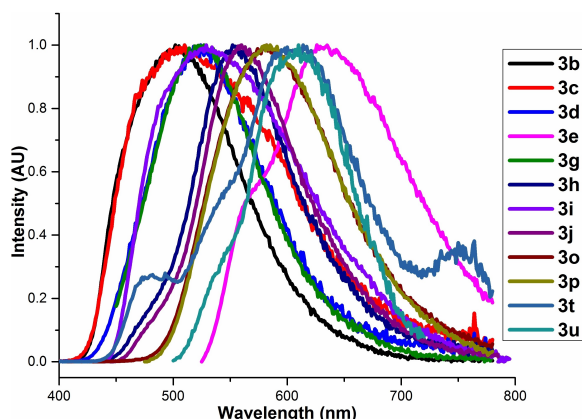


Figure 4. Normalized emission spectra of Knoevenagel adducts (3) in the solid state.

Stokes shift than **3o** (α -naphthyl group, λ_{em} =581 nm, Stokes shift=11037 cm^{-1}). 2-((2-aminopyridin-3-yl)methylene)malononitrile (**3t**, λ_{em} =605 nm, Stokes shifts=10719 cm^{-1}) was slightly red shifted with a large Stokes shift as compared to 2-((1H-indol-3-yl)methylene)malononitrile (**3u**, λ_{em} =608 nm, Stokes shift=9663 cm^{-1}). From this study, it was concluded that the electron donating groups on aromatic/heteroaromatic ring of Knoevenagel products (3) displayed D- π -A-type molecular architectures. Further, it was worthy to mention that the electron withdrawing groups on aromatic/heteroaromatic ring of Knoevenagel adducts (3) did not exhibit any fluorescence properties. This might be due to the presence of electron accepting groups (A- π -A-type) at both the ends of the Knoevenagel adducts (3).

The absorption and emission of β -enaminones (**6**) were strongly influenced by the substituents on aromatic ring of β -enaminones (**6**) (Figure S34 (see Supporting Information) and Figure 5). For example, the electron withdrawing nitro group at *para* position of the phenyl ring of β -enaminone (**6f**) (4-NO₂ phenyl λ_{em} =620 nm) was bathochromically shifted with a large Stokes shift (10681 cm^{-1}) when compared to the other electron

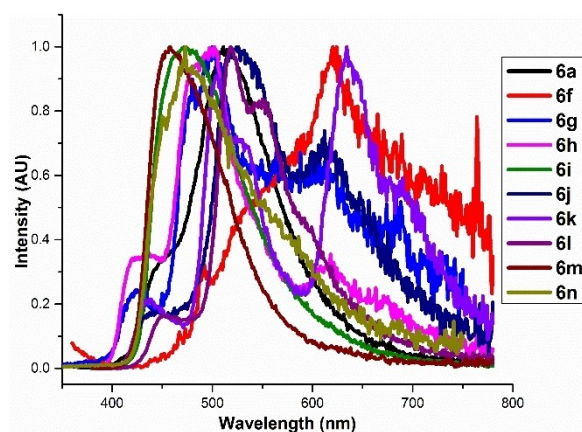


Figure 5. Normalized emission spectra of β -enaminones (6) in the solid state.

with drawing groups at different positions of the phenyl ring of β -enaminones i.e. **6g** (4-CN phenyl, λ_{em} =495 nm, Stokes shift=1971 cm^{-1}), **6h** (4-Cl phenyl, λ_{em} =491 nm, Stokes shift=8959 cm^{-1}), **6i** (4-Br phenyl, λ_{em} =472 nm, Stokes shift=8054 cm^{-1}) and **6j** (2, 4-dichloro phenyl, λ_{em} =526 nm, Stokes shifts=9891 cm^{-1}). The β -naphthylsubstituted β -enaminone, **6k** (λ_{em} =633 nm) was red shifted with a large Stokes shifts (13104 cm^{-1}) when compared with the phenyl substituted β -enaminone, **6a** (λ_{em} =512 nm, Stokes shift=10231 cm^{-1}). Pyridyl substituted β -enaminone, **6l** (λ_{em} =519 nm) was red shifted with a large Stokes shift (9303 cm^{-1}) as compared with thiophene and coumarin substituted β -enaminones, **6m** (λ_{em} =458 nm, Stokes shift=6819 cm^{-1}) and **6n** (λ_{em} =478 nm, Stokes shift=8930 cm^{-1}). The study reveals that the phenyl, phenyl with EWGs, β -naphthyl, pyridyl, thiophene and coumarin substituted β -enaminones exhibited D- π -A-type molecular architectures. Further, it was also noted that the EDGs on aromatic ring of β -enaminones (**6**) did not display any significant fluorescence properties. This might be due to the presence of EDGs (D- π -D-type) at both the ends of the β -enaminones (**6**).

Colour coordinates and CCT values of Knoevenagel adducts (3) and β -enaminones (6)

The Commission Internationale de l'Éclairage (CIE) colour coordinates (x, y) of the Knoevenagel adducts (3) and β -enaminones (6) were investigated from the respective emission spectra (Figure 4 and Figure 5) at different excitation wavelengths by following the reported procedure.^[57] The obtained results are presented in Table 6. The colour coordinates of Knoevenagel adducts (3) and β -enaminones (6) are shown in Figure 6 and Figure 7, respectively.

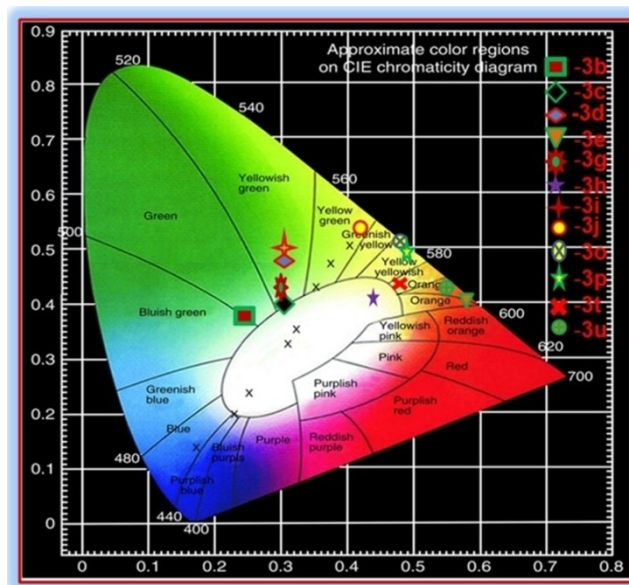


Figure 6. CIE chromaticity diagram of Knoevenagel adducts (3).

Table 6. Photophysical properties of Knoevenagel adducts (**3**) and β -enaminones (**6**).

Compd.	λ_{max} [nm] ^[a]	λ_{emi} [nm] ^[b]	Life time [μs] ^[c]	Stokes shift [cm ⁻¹] ^[d]	CIE Coordinates		CCT ^[e]
					X	Y	
3b	343	503	10.4	9274	0.24	0.39	9351
3c	303	507	8.8	13279	0.31	0.41	6224
3d	331	522	8.5	11054	0.31	0.49	–
3e	429	630	10.3	7437	0.58	0.41	–
3g	454	525	9.7	2979	0.30	0.42	–
3h	376	553	8.8	8513	0.44	0.41	3001
3i	367	529	9.2	8344	0.31	0.51	–
3j	355	561	11.3	10344	0.42	0.53	–
3o	354	581	8.4	11037	0.48	0.50	–
3p	365	585	9.6	10303	0.49	0.49	–
3t	367	605	8.6	10719	0.48	0.44	–
3u	383	608	10.5	9663	0.55	0.43	–
6a	336	512	9.4	10231	0.27	0.45	–
6f	373	620	9.4	10681	0.49	0.46	–
6g	451	495	7.7	1971	0.34	0.40	5270
6h	341	491	9.0	8959	0.25	0.34	10213
6i	342	472	10.0	8054	0.21	0.30	17380
6j	346	526	9.5	9891	0.38	0.48	–
6k	345	633	10.0	13104	0.38	0.44	4355
6l	350	519	10.0	9303	0.34	0.53	–
6m	349	458	10.0	6819	0.18	0.23	88932
6n	335	478	10.4	8930	0.25	0.33	10623

[a] Absorption measured in CH₃OH (1.0 $\times 10^{-5}$ M). [b] Emission measured in the solid state. [c] Life time in μs . [d] Stokes shift; $\Delta\nu = \nu_{\text{abs,max}} - \nu_{\text{emis,max}}$. [e] CCT = $-449n^3 + 3525n^2 - 6823n + 5520.33$; Here $n = (x-x_0)/(y-y_0)$, $x_0 = 0.3320$ and $y_0 = 0.1858$ (chromaticity epicenters). Cold white light CCT value is > 5000 K.

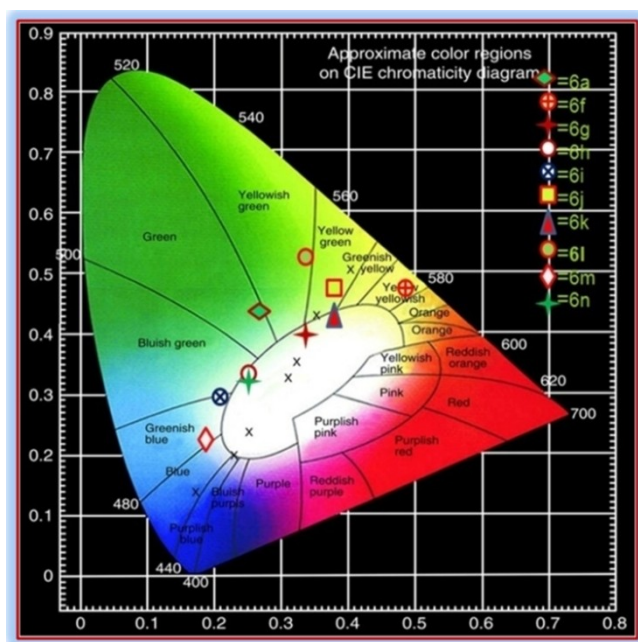


Figure 7. CIE chromaticity diagram of β -enaminones (**6**).

From CIE colour coordinates (x , y), it was found that the colour tunability of **3** and **6** depend mainly on EDGs and EWGs present on aromatic/heteroaromatic ring of **3** and **6**. For example, EDGs like 4-OMe (**3b**), 3-OH (**3c**), 4-OH (**3d**), 4-N,N-dimethyl (**3e**), 2,4-dihydroxy (**3g**), 2,4-diOMe (**3h**), 3-ethoxy-4-hydroxyphenyl (**3i**) & 3,4,5-triOMe (**3j**) groups on the aryl ring of Knoevenagel adducts (**3**) exhibited different colours on excitation at 410, 275, 275, 510, 405, 275, 425 & 430 nm,

respectively. As a result, compounds, **3b**, **3c**, **3d**, **3e**, **3g**, **3h**, **3i** and **3j** exhibited bluish-green (close to white border), cold white, yellowish-green, orange, yellowish-green, warm white, yellowish-green and greenish-yellow light emissions, respectively and were shown in terms of corresponding color coordinates as (0.24, 0.39), (0.31, 0.41), (0.31, 0.49), (0.58, 0.41), (0.30, 0.42), (0.44, 0.41), (0.31, 0.51) and (0.42, 0.53). It was worthy to note that the light emission was tunable from bluish-green to orange by the introduction of various EDG(s) at the different positions of aryl ring of Knoevenagel products (**3**). The Knoevenagel adducts, **3o** (α -naphthyl group, $\lambda_{\text{exi}} = 285$ nm, (0.48, 0.50)) and **3p** (β -naphthyl group, $\lambda_{\text{exi}} = 365$ nm, (0.49, 0.49)) showed yellow light emission. 2-((2-aminopyridin-3-yl)methylene)malononitrile (**3t**, $\lambda_{\text{exi}} = 340$ nm, (0.48, 0.44)) and 2-((1H-indol-3-yl)methylene)malononitrile (**3u**, $\lambda_{\text{exi}} = 450$ nm, (0.55, 0.43)) were emitted the yellow and orange light, respectively (Figure 6). Next, the EWGs like 4-NO₂ (**6f**), 4-CN (**6g**), 4-Cl (**6h**), 4-Br (**6i**) & 2,4-dichloro (**6j**) groups on the aryl ring of β -enaminones displayed various colours on excitation at 336, 451, 346, 347 & 351 nm, respectively. The study revealed that the compounds, **6f**, **6g**, **6h**, **6i** and **6j** showed yellow, warm white, cold white, cold white and yellow-green light emissions and are represented as (0.49, 0.46), (0.34, 0.40), (0.25, 0.34), (0.21, 0.30) and (0.38, 0.48) on the CIE colour coordinates, respectively. From this study, it was noticed that the light emission was tunable from cold white to yellow by the introduction of different EWGs at the different positions of aryl ring of β -enaminones(**6**). The phenyl and β -naphthyl substituted β -enaminones, **6a** and **6k** were excited at 344 and 349 nm exhibited green (0.27, 0.45) and warm white (0.38, 0.44) light emission, respectively. Interestingly, pyridyl, thiophene and coumarin substituted β -enaminones, **6l**, **6m** and **6n** excited at

354, 354 and 415 nm exhibited yellowish green (0.34, 0.53), cold white (0.18, 0.23) and cold white (0.25, 0.33) light emission, respectively (Figure 7).

The correlated colour temperature (CCT) values of **3** and **6** have been calculated. The obtained results are presented in Table 6. From the CCT values, it was observed that the compounds, **3b** (9351 K), **3c** (6224 K), **6h** (10213 K), **6i** (17380 K), **6m** (88932 K) & **6n** (10623 K) were emitted cold white light and **3h** (3001 K), **6g** (5270 K) & **6k** (4355 K) were emitted warm white light.

Fluorescent decay curves of Knoevenagel adducts (**3**) and β -enaminones (**6**)

To know the de-excitation dynamics of excited states and energy transfer between HOMO and LUMO levels of D- π -A-type of **3** and **6** (**3b**, **3c**, **3d**, **3e**, **3g**, **3h**, **3i**, **3j**, **3o**, **3p**, **3t** & **3u** and **6a**, **6f**, **6g**, **6h**, **6i**, **6j**, **6k**, **6l**, **6m** & **6n**), the fluorescent decay curves were recorded by monitoring emissions of the aforementioned compounds under the excitation wavelengths (Fig-

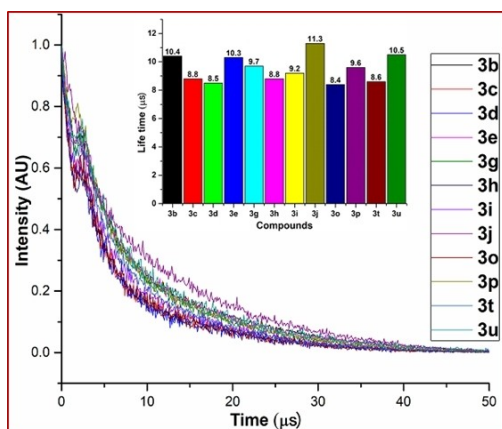


Figure 8. Decay curves of Knoevenagel adducts (**3**). Inset shows the lifetimes for the titled compounds.

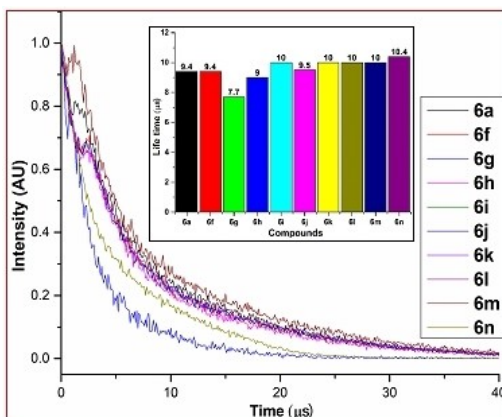


Figure 9. Decay curves of β -enaminones (**6**). Inset shows the lifetimes for the β -enaminones (**6**)

ure 8 and Figure 9). The decay curves of above compounds **3** and **6** showed a non-exponential nature owing to electrons transfer from donor to acceptor, therefore, the effective lifetime (τ_{eff}) has been evaluated using the following equation:

$$\tau_{eff} = \frac{\int I(t)dt}{\int I(t)dt} \quad (1)$$

Here $I(t)$ is the intensity of the luminescence at a given time t .

The lifetime values of compounds, **3b**, **3c**, **3d**, **3e**, **3g**, **3h**, **3i**, **3j**, **3o**, **3p**, **3t** & **3u** and **6a**, **6f**, **6g**, **6h**, **6i**, **6j**, **6k**, **6l**, **6m** & **6n** have been calculated by using Equation (1) (Table 6). As can be seen from Table 6, it was found that the lifetime values of compounds, **3b** (4-OMe), **3c** (3-OH), **3d** (4-OH), **3e** (4-N,N-dimethyl), **3g** (3,4-dihydroxy), **3h** (2,4-diOMe), **3i** (3-ethoxy-4-hydroxy) and **3j** (3,4,5-triOMe) containing EDGs on the phenyl ring of Knoevenagel adducts (**3**) ranging from 8.5 to 11.3 μ s. From this study, it was noticed that the compound, **3j** (3,4,5-trimethoxyphenyl, 11.3 μ s) has relatively greater lifetime as compared with other compounds, **3b**, **3c**, **3d**, **3e**, **3g**, **3h** and **3i**. The **3p** (β -naphthyl group, 9.6 μ s) has relatively longer lifetime than **3o** (α -naphthyl group, 8.4 μ s). 2-((1H-indol-3-yl)methylene)malononitrile (**3u**, 10.5 μ s) has relatively greater life time than 2-((2-aminopyridin-3-yl)methylene)malononitrile (**3t**, 8.6 μ s). Hence, the above Knoevenagel adducts spent lesser lifetimes at their excited states and displayed rapid radiative transition probabilities for highly efficient light emission. Further, the fluorescence intensity was calculated from the emission spectrum of respective Knoevenagel adducts (**3**). From the results, it was found that the emission intensity of compounds, **3c** (3-OH phenyl) and **3t** (2-aminopyridyl) were greater than the other compounds i.e. **3b**, **3d**, **3e**, **3g**, **3j**, **3o**, **3p**, **3t** and **3u** (Figure S34 (see Supporting Information)).

The lifetimes of β -enaminones, **6a**, **6f**, **6g**, **6h**, **6i**, **6j**, **6k**, **6l**, **6m** & **6n** were calculated and were ranged from 7.7 to 10.4 μ s (Table 6). The study revealed that the coumarin substituted β -enaminone, **6n** (10.4 μ s) has relatively greater lifetime as compared with other compounds, **6a**, **6f**, **6g**, **6h**, **6i**, **6j**, **6k**, **6l** & **6m**. Hence, the above β -enaminones also spent lesser lifetimes at their excited states and display rapid radiative transition probabilities for efficient light emission. The fluorescence intensity was calculated from the respective emission spectra of β -enaminones (**6**). The study reveals that the emission intensity of compounds containing strong EWGs on phenyl ring of β -enaminones, i.e. **6f** (4-NO₂ phenyl) and **6g** (4-CN phenyl) have comparatively higher than the other compounds (**6a** and **6h**-**6n**) (Figure S35 (see Supporting Information)).

Electrochemical properties

The electrochemical properties of D- π -A-type **3** & **6** have been studied by cyclic voltammetry (CV) (Figure 10, Figure S36 & S37(see Supporting Information) and Table 7). The CV measurements of **3** & **6** were taken in CH₃CN (0.5 mM) along with 0.1 M

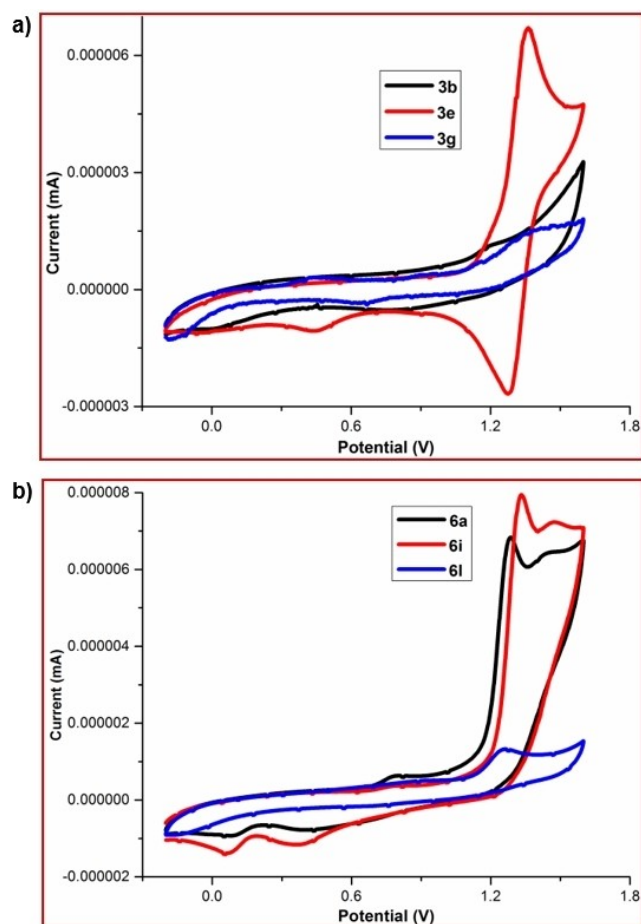


Figure 10. Cyclic voltammograms of a) Knoevenagel products (**3b**, **3e** & **3g**) and b) β -enaminones (**6a**, **6i** & **6l**) in CH_3CN (0.1 M Bu_4NPF_6 in CH_3CN) at a scan rate of 100 mVs^{-1} .

Table 7. Electrochemical data of Knoevenagel products (3) and β -enaminones (6).					
Compd.	λ_{edge} [nm]	$E_{\text{g}}^{\text{opt}}$ [eV] ^[a]	$E^{\text{ox}}_{\text{(onset)}}$	E_{HOMO} [eV] ^[b]	E_{LUMO} [eV] ^[c]
3b	388	3.19	1.096	−5.52	−2.33
3c	390	3.18	1.407	−5.83	−2.65
3d	386	3.21	1.289	−5.71	−2.50
3e	475	2.61	1.095	−5.51	−2.90
3g	500	2.48	1.044	−5.46	−2.98
3h	422	2.94	1.285	−5.70	−2.76
3i	416	2.98	1.212	−5.63	−2.65
3j	415	2.99	1.162	−5.58	−2.59
3o	415	2.99	1.133	−5.55	−2.56
3p	421	2.94	1.349	−5.77	−2.83
3t	399	3.11	1.372	−5.79	−2.68
3u	428	2.90	1.299	−5.72	−2.82
6a	388	3.19	1.084	−5.50	−2.31
6f	446	2.78	1.243	−5.66	−2.88
6h	393	3.15	1.155	−5.57	−2.42
6i	389	3.19	1.194	−5.61	−2.42
6j	397	3.12	1.237	−5.66	−2.54
6k	396	3.13	1.149	−5.57	−2.44
6l	408	3.04	1.088	−5.51	−2.47
6m	394	3.15	1.165	−5.58	−2.43
6n	385	3.22	1.167	−5.59	−2.37

[a] $E_{\text{g}}^{\text{opt}} = 1240/\lambda_{\text{edge}}$; [b] $E_{\text{HOMO}} = -[E_{\text{ox}}(\text{onset}) - E_{\text{FOC}} + 4.8]$ ($E_{\text{FOC}} = 0.38 \text{ eV}$ (FOC/FOC⁺ vs. Ag/AgCl)); [c] $E_{\text{LUMO}} = E_{\text{HOMO}} + E_{\text{g}}^{\text{opt}}$.

Bu_4NPF_6 (supporting electrolyte) in a standard 3-electrode system, reference electrode (Ag/Ag^+ (satd. KCl)), working electrode (platinum disk), counter electrode (platinum wire) and ferrocene was employed as an internal standard. All the CV scans were recorded in the potential range -0.2 to 1.6 V vs. Ag/AgCl (satd. KCl) with a scan rate of 100 mVs^{-1} . Cyclic voltammograms of **3** & **6** are shown in Figure 10. The HOMO and LUMO energy levels have been calculated and presented in Table 7. Most of the voltammograms of **3** & **6** were quasi-reversible in nature. Hence, the HOMO levels measured from onset potentials of the oxidation wave and the LUMO levels were from the HOMO levels and the ($E_{\text{g}}^{\text{opt}}$) (optical band gaps calculated from λ_{edge} of UV-Visible absorption spectrum) (Figure 10, Figure S36 & S37 (see Supporting Information) and Table 7). The HOMO/LUMO levels of **3** (**3b–3e**, **3g–3j**, **3o**, **3p**, **3t** & **3u**) and **6** (**6a**, **6f**, **6h**, **6i**, **6j**, **6k** & **6m**) were calculated and were noticed in the range of -5.83 to -5.46 eV / -2.98 to -2.33 eV and -5.66 to -5.50 eV / -2.88 to -2.31 eV . The HOMO levels of **3** (**3b** (-5.52 eV), **3c** (-5.83 eV), **3d** (-5.71 eV), **3h** (-5.70 eV), **3i** (-5.63 eV), **3j** (-5.58 eV), **3o** (-5.55 eV), **3p** (-5.77 eV), **3t** (-5.79 eV) & **3u** (-5.72 eV)) and **6** (**6a** (-5.50 eV), **6f** (-5.66 eV), **6h** (-5.57 eV), **6i** (-5.61 eV), **6j** (-5.66 eV), **6k** (-5.57 eV), **6l** (-5.51 eV), **6m** (-5.58 eV) & **6n** (-5.59 eV)) were comparable with the regularly employed HTMs.^[58–60] Based on the available literature reports, the ambipolar materials (APMs) have HOMO/LUMO energy levels that must be ≥ -5.6 / $\sim -3.15 \text{ eV}$.^[61] To our delight, the HOMO/LUMO levels of **3e** (-5.51 / -2.90 eV) and **3g** (-5.46 / -2.98 eV) were comparable to the most commonly employed APMs.^[62–65] Hence, the electrochemical behavior of most of the studied **3** and **6** are promising HTMs and APMs to made optoelectronic devices. Further, the lifetimes of HOMO/LUMO levels of **3** and **6** were calculated from luminescence decay curves and were ranging from 8.4 – $11.3 \mu\text{s}$ and 7.7 – $10.4 \mu\text{s}$, respectively. The decay curves of these levels of **3** and **6** displayed non-exponential behavior with lower lifetimes indicating a higher radiative transition probability which leads to efficient emission.

Conclusion

INDTNTs-4 employed as a task-specific heterogeneous green catalyst for carbon–carbon double bond forming reactions i.e. Knoevenagel condensation and β -enaminone formation using on hand starting materials under environmentally benign reaction conditions. Low catalyst loading, reusability of catalyst, easy to perform, broad substrate scope, no hazardous bases/solvents, high yields, and simple purification are the noteworthy advantages of the present green approach. Photophysical properties of the synthesized D- π -A-type Knoevenagel products (**3**) and β -enaminones (**6**) have been investigated. Considering the CIE color coordinates, it has been confirmed that the light emission in the solid state can be tuned from cold white light to orange light by the introduction of various electron donating and withdrawing group(s) at the different positions of aromatic/heteroaromatic ring of **3** and **6**, respectively. The HOMO levels of **3b**, **3c**, **3d**, **3h**, **3i**, **3j**, **3o**, **3p**, **3t** & **3u** and **6a**, **6f**, **6h**, **6i**, **6j**,

6k, **6l**, **6m** & **6n** were comparable with the normally used HTMs and the HOMO/LUMO energy levels of **3e** and **3g** were similar to the previously reported APMs. The lifetimes of HOMO/LUMO levels of **3** and **6** were calculated from luminescence decay curves and were ranging from 8.4–11.3 μ s and 7.7–10.4 μ s, respectively. The decay curves of these levels of **3** and **6** displayed non-exponential behavior with smaller lifetimes indicating a higher radiative transition probability which leads to efficient fluorescence. All the results of the synthesized small D- π -A-type **3** and **6** indicate that they are potential materials for the development of OLED devices.

Experimental Section

General information

The ^1H and ^{13}C NMR spectra were recorded on a Bruker NMR spectrometer (500 and 125 MHz). LC–MS spectra were recorded on Shimadzu-LCMS-2010 A mass spectrometer. XRD pattern was obtained from Rigaku Miniflex 600 diffractometer. High Resolution Transmission Electron microscopy (HRTEM) images obtained from FEI-Tecnai-F20 S-Twin at 200 kV. X-ray photoelectron spectra was recorded on ESCA LAB-250 Xi-XPS with $\text{Al}_{\text{K}\alpha}$ source. Absorbance spectra were recorded on Shimadzu model UV-3100 UV-Visible spectrophotometer. The progress of the reactions was monitored by TLC using 0.25 mm Merck silica gel plates and the spots were visualized under UV light. All chemicals and solvents were procured from Sigma Aldrich, Acros Organics Ltd., Merck etc. and used without any further purification.

Preparation of TNTs and INDNTs

TNTs and INDNTs were fabricated by following the reported procedure in the literature.^[48,50,51]

Hydrothermal synthesis of TNTs

The mixture of 3.0 g of anatase- TiO_2 (99%, Merck) and 100 mL of 10 M NaOH was taken in 250 mL beaker and was stirred for 1 h. Then the resulting solution was transferred into a Teflon-coated stainless steel (SS) autoclave, heated in a pre-heated electric oven at 150 °C for 24 h, and then cooled down to RT. The obtained colour less precipitate was washed several times with RO water, DD water and dilute HCl (0.1 M) followed by ethanol. The resulting TNTs were dried at 80 °C for 12 h.

Synthesis of INDNTs

The INDNTs were prepared using TNTs and β -alanine as nitrogen precursor. At first, a 1.0 mL of ethanol was added to each mixture of TNTs (0.4 g) and β -alanine (0.2, 0.4, 0.6 and 0.8 g) (Table 1). The each mixture was grounded well in a mortar-pestle separately and then transferred into different ceramic crucibles and calcined at 350 °C for 3–4 h (with 5 °C per minute heating rate) in a furnace, cooled down to RT and then washed with ethanol for 30 min. The each mixture was centrifuged at 10,000 rpm for 5–6 min and the obtained material was heated at 80 °C for 12 h. Finally, the obtained INDNTs were labeled as INDNTs-1, INDNTs-2, INDNTs-3 and INDNTs-4.

General procedure for the synthesis of Knoevenagel adducts (3):

A mixture of aromatic/hetero aromatic aldehyde (**1**) (5.0 mmol) and malononitrile (**2**) (5.0 mmol) and INDNTs-4 (20 mg) in ethanol was refluxed for 25–35 min. Reaction progress was monitored by TLC. Upon completion, ethyl acetate (10.0 mL) was added to the reaction mixture and the catalyst was removed by filtration. The organic phase was dried over anhydrous MgSO_4 and then concentrated under reduced pressure. The obtained crude product (**3**) was purified by recrystallization (Aq. ethanol).

General procedure for the synthesis of β -enaminones (6):

A mixture of aromatic/heteroaromatic ketones (**4**) (5.0 mmol) and N,N-dimethylformamide dimethyl acetal (**5**) (5.0 mmol) and INDNTs-4 (20 mg) was refluxed for 45–70 min. The progress of the reaction was monitored by TLC. Upon completion, ethyl acetate (10.0 mL) was added to the reaction mass and the catalyst was removed by filtration. The organic layer was dried over anhydrous MgSO_4 and then concentrated under reduced pressure. The obtained crude product (**6**) was purified by recrystallization (Aq. ethanol).

Deposition Number 2175753 (for **3b**) contains the supplementary crystallographic data for this paper. These data are provided free of charge by the joint Cambridge Crystallographic Data Centre and Fachinformationszentrum Karlsruhe Access Structures service.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

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